

MATH 11205: Machine Learning in Python 2025-2026

Project Description

We will be using a subset of the data collected by [UNICEF](#), called the Multiple Indicator Cluster Survey (MICS). A simplified version of the data will be used for this project, provided in the file `unicef_malawi.csv`, after some initial cleaning steps (described below). This smaller data set focuses only on data collected in Malawi in the years 2019-2020. The dataset collects a number of indicators on the well-being of children, including childhood depression. Your goal is to build a model to predict if children suffer from feelings of depression.

Assignment Goal

For the purpose of the project, consider yourself a **Data Science Consultant** who has been hired by UNICEF to analyse childhood depression in low-income countries. The mental health of the next generation – those aged under 18 years - is a societal priority. Identifying and treating mental health early in life has lifelong impacts on physical health, education, earning potential, relationships, identity formation and life satisfaction. Further, the burden of poor mental health disproportionately falls on lower- and middle-income settings (LMICs), and on women and young people in particular. Mental health is influenced by various factors, at the level of the child, parent, and societal environment. For an integrated model of mental health, an approach that combines these multiple factors together is required, and is possible through the UNICEF's Multiple Indicator Cluster Survey (MICS), which collects information on the child, parent, and household environment. For further details on the data, please see the main webpage for the [Multiple Indicator Cluster Survey \(MICS\)](#).

Towards this aim, you have been asked to use this data to build a classification model to predict if a child suffers from depressive feelings. Depression is a heterogeneous condition, and to the question *How often does the child seem very sad or depressed?*, survey responses vary from *never*, *a few times a year*, *monthly*, *weekly*, and *daily*. **You have been instructed to focus only on discriminating between no depression and any form of depression, combining all severities of depression into one category.** In addition, the organization is interested in **identifying important factors** that can have an impact on depression; this is useful not only to improve their understanding of the condition and the impact of inter-generational experiences and society, but also to assist in the development of effective social and health policies and interventions. In summary, you need to develop a **well-tuned and validated** classification model for **depression** as the binary outcome of interest and **interpret or explain** the model's predictions. Your model may use as few or as many of the provided features, transforming and manipulating these features in any way that you see fit, and you may extract additional information from the extended datasets to create new features.

You should start from an **interpretable baseline** model of your choice, including as few or as many of the provided variables. At this point, we have covered a number of models in lectures and workshops, and you may explore a variety of different feature engineering and modelling approaches for this particular task. However, your ultimate goal is to select and deliver a **single final model**. Thus, your report should focus on describing and motivating your final model choice, along with a comparison against the baseline model. It is important that any interpretations and conclusions you draw from your model are well supported and sound and that you understand limitations of the model and the data.

Working as a team

This project may be completed by a team of up to 4 students (minimum of 1 student). Feel free to create your own team during workshop hours, building on the pairs for the workshops. Since we are not assigning teams, if you are a team that is looking for more members or someone looking for a team please use the pinned post on Piazza to find each other. You are strongly encouraged to work in teams, as you can learn a lot from discussing together, but you may choose to work individually if preferred. **The marking scheme is the same, regardless of group size.**

After the assignment is completed, we will distribute a brief peer evaluation survey. Completion of the survey is optional, but if you feel that some members contributed significantly less, this provides an opportunity for feedback and for such members to potentially have their overall mark penalized. This will only be done in extreme cases, after discussion with all team members.

Required Structure

A Jupyter notebook template called `project.ipynb` has been provided. It includes the required sections along with brief instructions on what should be included in each section. Your completed assignment must follow this structure - **you should not add or remove any of these sections, but you may add extra subsections to help organize the report.** Please remove the instructions for each section in the final document.

All of your work must be contained in the `project.ipynb` notebook, we will only mark what is included in this file (both the write-up and relevant coding). You may work on the notebook in whichever environment you prefer (noteable, locally, colab, codespaces,...).

There is an **upper limit of 30 pages** including all code and output. Your notebook must include all of your work, but make sure that you are only retaining required components, e.g. remove unused code and figures (if a figure is not explicitly discussed in the text it should not be in the final document). Overall, your project will be partially assessed on your organization / presentation of the document - it should be as polished and streamlined as possible. **Try to be as concise as possible while creating your write-up. We highly recommend that you check the appearance of your rendered PDF before submitting, as its appearance can differ significantly from the notebook.**

Please submit your final PDF of project report (generated from a Jupyter notebook) to the Project assignment on Gradescope. Please ensure that you **tag all groups members** on Gradescope, and also add all group member names at the beginning of the file.

Getting Help

- **Week 11 Workshop:** We will focus on answering any project related questions.
- **Piazza:** This forum will be used as the central location for all course related discussions and questions, and should be used over emailing course staff directly. The course lecturer will monitor and respond to questions, but feel free to provide some constructive responses to peer's questions.

Generative AI Policy

Please refer to the University's [Generative AI Policy](#). Generative AI tools may be used for project, but always as your helper or co-pilot and used responsibly, and NOT your driver! For example, GenAI may be used for generating explanations for error messages and debugging, providing hints or suggestions to improve code, enhancing visualizations and the quality of the report. The data should absolutely NOT be copied and shared directly with GenAI tools, as this is in breach of legal consent for this data. You must include a statement on how GenAI was used. **If the project is suspected to be heavily written by GenAI, the students may be subject to an oral presentation.**

Dataset Details

The data provided in `unicef_malawi.csv` merges together a subset of variables from the questionnaires on the child, mother, and household.

Child-level variables: from the original data (provided in `ExtendedDataSources/fs.sav`), a subset of variables have been extracted:

- **Child's background:**
 - age, education, and health insurance coverage information are provided in CB3, CB4, CB5A, CB5B, CB7, CB11, with detailed questions and response options in pg. 2 of `Questionnaires/MICS6-Questionnaire_for_Children_Age_5_17.docx`
 - region of residence, gender, ethnicity, and combined wealth score are provided in HH6, HH7, HL4, ethnicity, wscore.
- **Child labour:**
 - child labour, hours of labour, household work, and hours of household work are provided in CL2, CL3, CL12, CL13, with detailed questions and response options in pg. 4-6 of `Questionnaires/MICS6-Questionnaire_for_Children_Age_5_17.docx`
- **Child discipline:**
 - discipline methods and physical punishment are provided in FCD2A, FCD2B, FCD2C, FCD2D, FCD2E, FCD2F, FCD2G, FCD2H, FCD2I, FCD2J, FCD2K, FCD5, with detailed questions and response options in pg. 7 of `Questionnaires/MICS6-Questionnaire_for_Children_Age_5_17.docx`
- **Child functioning:**
 - **depression**, our target variable, is provided in FCF26, with detailed questions and response options in pg. 10 of `Questionnaires/MICS6-Questionnaire_for_Children_Age_5_17.docx`

Maternal variables: from the original data (provided in `ExtendedDataSources/wm.sav`), a subset of variables have been extracted:

- **Mother's background:**
 - age, education, and literacy are provided in WB4, WB5, WB6A, WB6B, WB14, with detailed questions and response options in pg. 2-3 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Domestic violence:**
 - attitudes towards domestic violence are provided in DV1A, DV1B, DV1C, DV1D, DV1E, with detailed questions and response options in pg. 29 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Victimization:**
 - information on attacks, harassment, and safety are provided in VT1, VT9, VT20, VT21, VT22A, VT22B, VT22C, VT22D, VT22E, VT22F, VT22X, with detailed questions and response options in pg. 30-32 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Marriage/Union:**
 - marital status is provided in 'MSTATUS'.
 - husband's age and multiple wives are provided in MA2, MA3, with detailed questions and response options in pg. 33 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Adult functioning:**
 - disability is provided in 'disability'.
 - difficulty in remembering, self-care, and communicating are provided in 'AF10', 'AF11', 'AF12', with detailed questions and response options in pg. 34 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Tobacco and alcohol use:**
 - tobacco and alcohol use are provided in 'TA1', 'TA14', with detailed questions and response options in pg. 43-44 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Life satisfaction:**
 - life satisfaction information is provided in 'LS1', 'LS2', 'LS3', 'LS4', with detailed questions and response options in pg. 45 of `Questionnaires/MICS6-Questionnaire_for_Individual_Women.docx`
- **Fertility:**
 - the number of surviving and deceased children is provided in 'CSURV', 'CDEAD'.

Household variables: from the original data (provided in `ExtendedDataSources/hh.sav`), a subset of variables have been extracted:

- **Household characteristics:**
 - house material, electronic use, ownership (dwelling, land, and/or livestock), and bank account possession information are provided in 'HC4', 'HC5', 'HC8', 'HC11', 'HC12', 'HC13', 'HC14', 'HC15', 'HC17', 'HC19', with detailed questions and response options in pg. 5-8 of `Questionnaires/MICS6-Household-Questionnaire.docx`
- **Insecticide treated nets:**
 - mosquito net information is provided in 'TN1', with detailed questions and response options in pg. 13 of

Questionnaires/MICS6_Household_Questionnaire.docx

- **Water and sanitation:**

- source, location, and sufficiency of drinking water and toilet facilities information are provided in 'WS1', 'WS3', 'WS4', 'WS7', 'WS11', 'WS14', 'WS15', with detailed questions and response options in pg. 15-18 of

Questionnaires/MICS6_Household_Questionnaire.docx

- **Handwashing:**

- availability of soap/detergent for handwashing is provided in 'HW5', with detailed questions and response options in pg. 19 of

Questionnaires/MICS6_Household_Questionnaire.docx

For further details on the variables, questions, and responses, please see the **Questionnaires** folder provided in project materials. As mentioned, **the target variable is depression in the child FCF26, which you need to binarize into the two categories 'no depression' and 'depression'** (combining all severities of depression into one category). To predict depression, you may choose to include as few or as many of the variables. You may also include additional variables from the extended datasets provided in the folder **ExtendedDataSources**. In this case, please describe the additional variables and your motivation for including them.

The data provided for this project is subject to data agreements and waivers that I have signed on your behalf. Thus, **the data must NOT be shared publicly** (e.g. if your team is using GitHub, keep the repo private and do NOT share the data with any GenAI tools).